Yang Jason Xiao, Ph.D.

Assistant Professor of Chemical Engineering
Principal Investigator, *Reaction Engineering and Catalysis Science Laboratory (RECSL)*
Department of Chemical Engineering, Institute for Micromanufacturing (IfM)
Louisiana Tech University, 505 Tech Drive, Ruston, LA 71272
E-mail: yxiao@latech.edu; Office Tel.: 318-257-5109; Fax: 318-257-4000
Group Website: http://www2.latech.edu/~yxiao/

Education and Professional Preparation

Postdoctoral Research Associate **Purdue University** (West Lafayette, IN, U.S.A.) 2015 - 2018
- *Catalytic reaction engineering with Prof. Arvind Varma*
- *Computational catalysis with Prof. Jeffrey P. Greeley*

Ph.D. in Chemical Engineering **Purdue University/Southeast University** 2015
Ph.D. Co-advisors:
- *Prof. Arvind Varma, Purdue University 2012 - 2015*
- *Prof. Guomin Xiao, Southeast University 2010 - 2012*

B.S. in Chemical Engineering **Southeast University** (Nanjing, China) 2010

Principal Research Interests

Chemical Reaction Engineering, Catalysis Science, Process Engineering, and Multiscale Modeling.

Appointments

Assistant Professor	Louisiana Tech University	2022 - Present
Senior Research Engineer	Purdue University	2019 - 2022
Research Scientist	Purdue University	2018 - 2019

Awards and Recognitions

Virgil Orr Junior Faculty Award	Louisiana Tech University (LaTech)	2026
College Education Award	College of Engineering and Science, LaTech	2026
College Research Award: DARPA SMS	College of Engineering and Science, LaTech	2026
College Research Award: DARPA MICA	College of Engineering and Science, LaTech	2026
Rising Star in Sustainability	Sustainability Science and Technology	2026
L. K. Doraiswamy Graduate Seminar Series	Iowa State University (ISU)	2026
College Research Award	College of Engineering and Science, LaTech	2025
Early Career Excellence in Research Award	Louisiana Tech University (LaTech)	2025
Doctoral New Investigator Award	ACS Petroleum Research Fund (ACS-PRF)	2025
Engineering Research Initiation (ERI) Award	U.S. National Science Foundation (NSF)	2024
Research Enhancement Awards	Louisiana Space Grant (LaSPACE)	2023
Young Investigator Travel Award	Intl. Symp. Chem. React. Eng. (ISCRE)	2023
Collaborative Partnership Awards	Louisiana Materials Design Alliance	2023

Yang Jason Xiao, Ph.D. Web: http://www2.latech.edu/~yxiao/
Assistant Professor, Louisiana Tech University E-mail: yxiao@latech.edu

Research Grants as PI/Co-PI at Louisiana Tech University

18. "Predictive Safety Modeling of Reactive Particles in Space Manufacturing and In-Situ Resource Utilization"
Louisiana Space Grant Consortium, *LURA* $4,500 2026 - Present

17. "Selective Semi-Hydrogenation of Acetylene to Ethylene over Two-Dimensional Nanolayer Catalysts"
ACS Petroleum Research Fund, *69076-DNI5* $110,000 2025 - Present

16. "Photo-Thermo Catalysis for Converting CO_2 to Ethanol"
Louisiana Materials Design Alliance, *SURE* $5,000 2025 - Present

15. "Clean Hydrogen from Methane for Propulsion Engines"
Louisiana Space Grant Consortium, *LURA* $4,500 2025 - 2026

14. "Elucidation of the Active Site for Hydrogen Generation from Methane over MXene Catalysts"
U.S. Department of Energy, *SCGSR* $43,200 2025 - 2026

13. "First-Principles and Machine Learning Investigations of High-Entropy MXene Catalysts"
Louisiana Materials Design Alliance, *Track 1B* $40,000 2025 - 2026

12. "Adaptive Multi-Physics Engine for Reaction Engineering"
U.S. Defense Advanced Research Projects Agency, *MICA* $1,912,000 2025 - Present

11. "Crowded, Heterogeneous, Intracellular, and Multi-Scale Environments for Revolutionary Bio-Applications"
U.S. Defense Advanced Research Projects Agency, *SMS* $1,251,691 2025 - Present

10. "Presenting Louisiana Tech University's NASA-Related Energy Research at NASCRE-5"
Louisiana Board of Regents, *TAP* $2,500 2025

9. "Understanding the Stability of MXene-Confined Nanolayer Catalysts for Ethane Dehydrogenation"
U.S. National Science Foundation, *CBET #2414204/#2414205* $501,766 2024 - Present

8. "Advancing Catalytic Conversion of Methane to Carbon-Free Hydrogen and Ethane"
U.S. National Science Foundation, *CBET #2347475* $199,954 2024 - Present

7. "Computational Investigation of the Stability of High-Entropy MXene Catalysts"
Louisiana Materials Design Alliance, *Track 1B* $40,000 2024 - 2025

6. "Photo-Thermo Catalysis Enhanced Oligomerization of Olefins to Liquid Rocket Propellant Fuels"
Louisiana Space Grant Consortium, *REA* $70,000 2024 - 2025

5. "Conversion of Methane to Transportation Fuels via Photo-Thermo Catalysis"
Louisiana Transportation Research Center, *TIRE* $30,000 2024 - 2025

4. "Characterization of Catalysts using Time-of-Flight Secondary Ion Mass Spectrometry"
Louisiana Materials Design Alliance, *LINK* $7,000 2024

3. "Developing Structure-Property Relation of HE-MXene Catalysts Using Machine Learning"
Louisiana Materials Design Alliance, *SURE* $5,000 2024 - 2025

2. "Design Catalytically Active Sites at High-Entropy MXene and Metallic Nanolayer Interfaces"
Louisiana Materials Design Alliance, *Track 1B* $39,953 2023 - 2024

1. "Catalytic Conversion of Methane to Liquid Rocket Propellant Fuels"
Louisiana Space Grant Consortium, *REA* $70,000 2023 - 2024

Research Group at Louisiana Tech University

Postdoctoral Research Associates:

Dr. Jake Vu *2026 - Present*
Dr. Delong Duan *2026 - Present*

Ph.D. Students:

Tobias K. Misicko (Ph.D. candidate) *2022 - Present*

Recipient of the 2025 DOE Office of Science Graduate Student Research (SCGSR) Fellowship

Joaquin Herrero *2024 - Present*
Jiaping Weng *2024 - Present*
Jacob Robinson *2024 - Present*
Keshar Sanjel (co-advised) *2025 - Present*

Undergraduate Students:

Piper Smith *2023 - Present*

Recipient of LaSPACE Undergraduate Research Assistantship (LURA) Award

Andres Rodriguez *2024 - Present*
Natalie Rayne Remedies *2025 - Present*
Charles Mitchell *2026 - Present*
Kohl Garner *2026 - Present*
Isabelle Lieux *2026 - Present*

Recipient of LaSPACE Undergraduate Research Assistantship (LURA) Award

Group Alumni

Master's Students:

Henrik Ketting *2023 - 2024*
Gabriel Chukwuka *(co-advised)* *2023 - 2025*

Undergraduate Students:

Caroline Cresap *2023 - 2026*

Recipient of LAMDA - Supervised Undergraduate Research Experiences (SURE) Award

Kaleigh Louque *2024 - 2026*

Recipient of the 2025 DOE Science Undergraduate Laboratory Internships (SULI)

Katherine Cortez *2024 - 2026*
Brady Duplessis *2022 - 2023*
Sean Clay *2023 - 2024*
Sarah Siharath *2023 - 2024*
Gregory Allen *2023 - 2024*
Robert Martin *2023 - 2025*

High School Students:

Piper Smith	*2023*
Mir Z. Ali	*2023*

Teaching Experience

Chemical Plant Design I (CMEN 431)	Louisiana Tech University	2024 - Present
Chemical Plant Design II (CMEN 432)	Louisiana Tech University	2024 - Present
Chemical Plant Design III (CMEN 434)	Louisiana Tech University	2025 - Present
Sustainable Chemical Processes (CMEN 450C/557)	Louisiana Tech University	2022 - 2024
Transport Phenomena (CMEN 304)	Louisiana Tech University	2022 - 2024
Engineering Problem Solving II (ENGR 121)	Louisiana Tech University	2022 - 2023
Heat and Mass Transfer (CHE 378)	Purdue University	2021
Chemical Reaction Engineering (CHE 660)	Purdue University	2019

Senior Design / Capstone Project Supervision (CMEN 431, CMEN 432, and CMEN 434):

•2026 - 2027: 23 students; two capstone design projects

(1) Acrylonitrile Production by Propylene Ammoxidation (SOHIO-Type Process)

(2) Continuous Production of p-Chloroaniline by Selective Hydrogenation of p-Chloronitrobenzene

•2025 - 2026: 36 students; three capstone design projects

(1) Closing the Carbon Loop: CO_2 Capture and Catalytic Hydrogenolysis to Methanol

(2) Fuel-Octane Booster: MTBE Synthesis Using Catalytic Distillation

(3) Materials for Modern Life: Ethylene Glycol Production Design

•2024 - 2025: 25 students; five capstone design projects

(1) Cumene Production from an Aromatics Complex

(2) Hydrogen and Olefins from Louisiana Shale Gas

(3) Hydrocracking Process Design for Superior Fuels and Lubricants Production

(4) FCC Gasoline Desulfurization

(5) Methanol to Olefins (MTO) Process

Peer-Reviewed Book Chapter

[1] Jacob C. Robinson, Jiaping Weng, Tobias K. Misicko, Daniela S. Mainardi, and Yang Xiao*, "Selective Passivized Catalysis: Balancing Activity, Selectivity, and Stability," *RSC Catalysis Series*, Vol. 38, Royal Society of Chemistry, **2026**, ISSN 1757-6725.

Peer-Reviewed Journal Publications

$*$ *corresponding authors;* † *these authors contributed equally*

39. Tobias K. Misicko†, Xiaopeng Liu†, Yue Wu*, and Yang J. Xiao*, Selective Passivized Catalysis in Nonoxidative Coupling of Methane over Pt Nanolayer/Mo_2TiC_2 MXene: Insights from Reaction–Diffusion

Analysis and Active Site Identification, *under review*, **2026**.

38. Joaquin Herrero, Tobias K. Misicko, Vijaya Gopu, and Yang Xiao*, Selective Hydrogenolysis of Glycerol to 1,3-Propanediol over Bifunctional Pt/KIT-6 Catalysts, *Sustainability Science and Technology*, **2026**, 3 (2), 024006.
37. Tobias K. Misicko†, Joaquin Herrero†, Jacob C. Robinson†, Daniela S. Mainardi, and Yang Xiao*, Thermal Runaway of Chemical Reactors: An Experimental, Modeling, and Machine-Learning Investigation, *Industrial & Engineering Chemistry Research*, **2026**, 65 (4), 2057-2070.
36. Jacob C. Robinson†, Jiaping Weng†, Tobias K. Misicko†, Natalie Remedies, Daniela S. Mainardi, and Yang Xiao*, Non-oxidative Coupling of Methane via Selective Passivized Catalysis *(Feature Article, Cover Art)*, *Chemical Communications*, **2025**, 61 (97), 19183-19194.
35. Gabriel Parker†, Tobias K. Misicko†, Tanguy Teriler, Yang Xiao, and Xiao-Ying Yu*, ToF-SIMS Spectral Analysis of Pristine and Neutron Irradiated Single Crystal Tungsten, *Results in Surfaces and Interfaces*, **2025**, 20, 100577.
34. Joaquin Herrero†, Peilei He†, Fan Yang, Jiaping Weng, Nicole LiBretto, Daniela S. Mainardi, Jeffrey T. Miller*, Yue Wu*, and Yang Xiao*, Synergy of Cu(I) and Oxygen Vacancies in CO_2 Hydrogenative Coupling to Ethanol on Cu/CeO_{2-x} Catalysts, *Nano Research*, **2025**, 18 (8), 94907518.
33. Rui Zhou, Mingzhou Jin*, Zhenglong Li, Yang Xiao, David McCollum, and Alicia Li, Techno-economic Analysis and Network Design for CO_2 Conversion to Jet Fuels in the United States, *Renewable and Sustainable Energy Reviews*, **2025**, 210, 115191.
32. Hao Li, Xue Liu, Yang Xiao*, and Kun Cao*, One-pot Synthesis of 5-Norbornene-2,3-Dicarboxylic Anhydride with High Exo/Endo Ratio in a Microreactor under High Temperature and Appropriate Pressure, *Chemical Engineering Journal*, **2024**, 498, 155561.
31. Hao Li, Xue Liu, Yan Zhang, Yang Xiao*, and Kun Cao*, A Kinetic Model of the Cycloaddition Reactions Between Cyclopentadiene and 1,3-Butadiene for Synthesis of 5-vinyl-2-norbornene, *The Canadian Journal of Chemical Engineering*, **2024**, 102 (5), 1946-1956.
30. Zhe Li†, Tobias K. Misicko†, Fan Yang†, Xiaopeng Liu, Zhenwei Wu, Xiaoyang Gao, Tao Ma, Jeffrey T. Miller, Daniela S. Mainardi, Collin D. Wick, Zhenhua Zeng*, Yang Xiao*, and Yue Wu*, Two-dimensional Atomically Thin Pt Layers on MXenes: The Role of Electronic Effects During Catalytic Dehydrogenation of Ethane and Propane, *Nano Research*, **2024**, 17 (3), 1251-1258.
29. Zhichao Shang, Teng Wang, Aoxia Ren, Yong Yu, Yan Zheng, Yuan Tao, Peizhong Feng, Yang Xiao*, and Xiaohong Wang*, Hollow Macroporous CeO_2/β-Bi_2O_3 Heterostructure Sphere via One-step Spray Solution Combustion Synthesis for Efficient Photocatalysis, *Applied Surface Science*, **2023**, 619, 156718.
28. Zhichao Shang, Yong Yu, Hang Yang, Zhongxiang Yang, Yang Xiao*, and Xiaohong Wang*, One-step Solution Combustion Synthesis of Micro/nano-scale Porous Cu/CeO_2 with Enhanced Photocatalytic Properties, *Journal of Rare Earths*, **2023**, 41 (2), 250-258.
27. Yang Xiao*, Anand Ramanathan, Bala Subramaniam*, and Arvind Varma, Guaiacol Hydrodeoxygenation and Hydrogenation over Bimetallic Pt-M (Nb, W, Zr)/KIT-6 Catalysts with Tunable Acidity, *ACS Sustainable Chemistry & Engineering*, **2022**, 10 (15), 4831-4838.

26. Zhe Li†, Yang Xiao*†, Prabudhya Roy Chowdhury, Zhenwei Wu, Tao Ma, Johnny ZhuChen, Gang Wan, Tae-Hoon Kim, Dapeng Jing, Peilei He, Pratik J. Potdar, Lin Zhou, Zhenhua Zeng, Xiulin Ruan, Jeffrey T. Miller, Jeffrey P. Greeley, Yue Wu*, and Arvind Varma, Direct Methane Activation by Atomically Thin Platinum Nanolayers on Two-Dimensional Metal Carbides, *Nature Catalysis*, **2021**, 4, 882-891.
25. Ruiping Wei, Xumin Qu, Yang Xiao*, Jingdeng Fan, Gaoli Geng, Lijing Gao, and Guomin Xiao*, Hydrogenolysis of Glycerol to Propanediols over Silicotungstic Acid Catalysts Intercalated with CuZnFe Hydrotalcite-like Compounds, *Catalysis Today*, **2021**, 368, 224-231.
24. Chunxiao Xu, Yayong Li, Ryan A. Adams, Vilas G. Pol, Yang Xiao*, Arvind Varma, and Pengwan Chen*, One-Step Combustion Synthesis of Carbon-Coated NiO/Ni Composites for Lithium and Sodium Storage, *Journal of Alloys and Compounds*, **2021**, 884, 160927.
23. Yuanfeng Wu, Yang Xiao, Hui Yuan, Zongqi Zhang, Shengbin Shi, Ruiping Wei, Lijing Gao, and Guomin Xiao*, Imidazolium Ionic Liquid Functionalized UiO-66-NH_2 as Highly Efficient Catalysts for Chemical Fixation of CO_2 into Cyclic Carbonates, *Microporous and Mesoporous Materials*, **2021**, 310, 110578.
22. Feng Jiang, Shanshan Wang, Bing Liu, Jie Liu, Li Wang, Yang Xiao, Yuebing Xu, and Xiaohao Liu*, Insights into the Influence of CeO_2 Crystal Facet on CO_2 Hydrogenation to Methanol over Pd/CeO_2 Catalysts, *ACS Catalysis*, **2020**, 10 (19), 11493-11509.
21. Zhichao Shang, Zhongxiang Yang, Yang Xiao*, and Xiaohong Wang*, Ordered Mesoporous Ag/CeO_2 Nanocrystalline via Silica-templated Solution Combustion for Enhanced Photocatalytic Performance, *Colloids and Surfaces A: Physicochemical and Engineering Aspects*, **2020**, 604, 125301.
20. Yang Xiao*, Rexonni Lagare, Lindsey Blanshan, Enrico N. Martinez, and Arvind Varma, Refinement of the Kinetic Model for Guaiacol Hydrodeoxygenation over Platinum Catalysts, *AIChE Journal*, **2020**, 66 (5), e16913.
19. Kanak Roy, Luca Artiglia, Yang Xiao, Arvind Varma*, and Jeroen A. van Bokhoven*, Role of Bismuth in the Stability of Pt-Bi Bimetallic Catalyst for Methane Mediated Deoxygenation of Guaiacol, an APXPS Study, *ACS Catalysis*, **2019**, 9 (4), 3694-3699.
18. Johnny ZhuChen, Zhenwei Wu, Xiaoben Zhang, Slgi Choi, Yang Xiao, Arvind Varma, Wei Liu, Guanghui Zhang*, and Jeffrey T. Miller*, Identification of the Structure of the Bi Promoted Pt Non-oxidative Coupling of Methane Catalyst: A Nanoscale Pt_3Bi Intermetallic Alloy, *Catalysis Science & Technology*, **2019**, 9, 1349-1356.
17. Yuan Wang, Yang Xiao*, and Guomin Xiao, Sustainable Value-added C_3 Chemicals from Glycerol Transformations: A Mini Review for Heterogeneous Catalytic Processes, *Chinese Journal of Chemical Engineering*, **2019**, 27, 1536-1542.
16. Xiaoni Du, Xiaohong Wang*, and Yang Xiao, $Bi_5O_7NO_3$ and Ag/$Bi_5O_7NO_3$ Composites: One-step Solution Combustion Synthesis, Characterization and Photocatalytic Properties, *CrystEngComm*, **2018**, 20, 7536-7542.
15. Yang Xiao, Yuan Wang, and Arvind Varma*, Low-Temperature Selective Oxidation of Methanol over Pt-Bi Bimetallic Catalysts, *Journal of Catalysis*, **2018**, 363, 144-153.

14. Yang Xiao and Arvind Varma*, Highly Selective Nonoxidative Coupling of Methane over Pt-Bi Bimetallic Catalysts, *ACS Catalysis*, **2018**, 8 (4), 2735-2740.
13. Yang Xiao and Arvind Varma*, Bio-oil Upgrading Using Methane: A Mechanistic Study of Model Compound Guaiacol Reactions over Pt-Bi Bimetallic Catalysts, *ACS Sustainable Chemistry & Engineering*, **2018**, 6 (12), 17368-17375.
12. Yang Xiao and Arvind Varma*, Kinetics of Glycerol Conversion to Hydrocarbon Fuels over Pd/H-ZSM-5 Catalyst, *AIChE Journal*, **2017**, 63 (12), 5445-5451.
11. Yang Xiao and Arvind Varma*, Kinetics of Guaiacol Deoxygenation over the Pt-Bi Catalyst, *Reaction Chemistry & Engineering*, **2017**, 2, 36-43.
10. Yang Xiao, Jeffrey Greeley, Arvind Varma*, Zhi-Jian Zhao, and Guomin Xiao, An Experimental and Theoretical Study of Glycerol Oxidation to 1,3-Dihydroxyacetone Over Bimetallic Pt-Bi Catalysts, *AIChE Journal*, **2017**, 63 (2), 705-715. **(Most Read Article)**
9. Yang Xiao and Arvind Varma*, Conversion of Glycerol to Hydrocarbon Fuels via Bifunctional Catalysts, *ACS Energy Letters*, **2016**, 1 (5), 963-968.
8. Danni Gao, Yang Xiao, and Arvind Varma*, Guaiacol Hydrodeoxygenation over Platinum Catalyst: Reaction Pathways and Kinetics, *Industrial & Engineering Chemistry Research*, **2015**, 54 (43), 10638-10644.
7. Yang Xiao and Arvind Varma*, Catalytic Deoxygenation of Guaiacol Using Methane, *ACS Sustainable Chemistry & Engineering*, **2015**, 3 (11), 2606-2610.
6. Haoyang Li, Xiaomei Pan, Yang Xiao, Guomin Xiao*, and Jinjin Huang, Simulation of Biodiesel Industrial Production via Solid Base Catalyst in a Fixed-bed Reactor, *Journal of Southeast University (English Edition)*, **2014**, 30 (3), 380-386.
5. Yang Xiao, Guomin Xiao, and Arvind Varma*, A Universal Procedure for Crude Glycerol Purification from Different Feedstocks in Biodiesel Production: Experimental and Simulation Study, *Industrial & Engineering Chemistry Research*, **2013**, 52 (39), 14291-14296.
4. Yang Xiao, Haoyang Li, Guomin Xiao*, Lijing Gao, and Xiaomei Pan, Simulation of the Catalytic Reactive Distillation Process for Biodiesel Production via Transesterification, *International Conference on Materials for Renewable Energy and Environment*, Chengdu, China, **2013**, 1, 196-199.
3. Guomin Xiao*, Haoyang Li, and Yang Xiao, Applications of Chemical Simulation Softwares in the Course of Transport Phenomena, *Chemical Industry Times*, **2012**, 26 (10), 53-59.
2. Yang Xiao, Lijing Gao, Guomin Xiao*, Baosong Fu, and Lei Niu, Experimental and Modeling Study of Continuous Catalytic Transesterification to Biodiesel in a Bench-Scale Fixed-Bed Reactor, *Industrial & Engineering Chemistry Research*, **2012**, 51 (37), 11860-11865.
1. Yang Xiao, Lijing Gao, Guomin Xiao*, and Jianhua Lv, Kinetics of the Transesterification Reaction Catalyzed by Solid Base in a Fixed-bed Reactor, *Energy & Fuels*, **2010**, 24 (11), 5829-5833.

U.S. Patents and Patent Applications

Granted U.S. Patents

7. Yang Xiao, Zhe Li, Yue Wu, and Fan Yang, Transition Metal Carbides for Catalytic Dehydrogenation of Short Alkanes, Patent No. US12698246B2, August 4, **2026**.
6. Nien-Hwa Linda Wang, Kai Jin, Clayton Gentilcore, and Yang Xiao, Integrated Continuous Conversion and Separation Methods for Upcycling Mixed Plastic Waste to Clean Gasoline and Diesel Fuels and Other Products, Patent No. US12516248B2, January 6, **2026**.
5. Yang Xiao, Arvind Varma, Zhe Li, and Yue Wu, Transition Metal Carbides for Catalytic Methane Activation, Patent No. US11524279B1, December 13, **2022**.
4. Yang Xiao and Arvind Varma, Method of Conversion of Glycerol to Hydrocarbon Fuels, Patent No. US10781376B2, September 22, **2020**.
3. Yang Xiao and Arvind Varma, Method of Producing Hydrocarbons from Methane, Patent No. US10450247B2, October 22, **2019**.
2. Yang Xiao and Arvind Varma, Method of Producing Formaldehyde from Methanol, Patent No. US10392333B2, August 27, **2019**.
1. Yang Xiao and Arvind Varma, Catalytic Deoxygenation of Bio-Oils Using Methane, Patent No. US10023809B2, July 17, **2018**.

U.S. Patent Applications, Provisional Patents, and Patent Disclosures

5. Yang Xiao, Tobias K. Misicko, Joaquin Herrero, Jacob C. Robinson, and Daniela S. Mainardi, Generalized Criterion for Quantification of Chemical Reactor Runaway Risk, Provisional Patent, January 10, **2026**.
4. Yang Xiao, Tobias K. Misicko, and Joaquin Herrero, Tunable Acidity of Pt/M KIT-6 Catalysts for the Purpose of Enhancing Hydrogenolysis of Glycerol to 1,3-Propanediol and Other Products, Provisional Patent, February 5, **2026**.
3. Yang Xiao, Yue Wu, Xiaopeng Liu, and Joaquin Herrero, Highly Selective Hydrogenation of Carbon Dioxide to Ethanol, Provisional Patent, August 20, **2025**.
2. Yang Xiao, Tobias K. Misicko, Xiaopeng Liu, and Yue Wu, Metal Carbide-based Catalyst Systems for Plastic Recycling, Patent Application 18/913,376, October 11, **2024**.
1. Yang Xiao, Arvind Varma, Anand Ramanathan, and Bala Subramaniam, Method of Enhanced Aromatic Products from Bio-oil Upgrading, Patent Application US20220356137A1, June 22, **2019**.

Invited Talks and Conference Presentations

58. Joaquin Herrero, Peilei He, Jiaping Weng, Daniela Mainardi, Jeffrey T. Miller, Yue Wu, and Yang Xiao*, CO_2 Hydrogenative Coupling to Ethanol on Cu/CeO_{2-x} Catalysts, *2026 AIChE Annual Meeting, Catalysis and Reaction Engineering Division*, Minneapolis, MN, November 8–12, **2026**.
57. Tobias K. Misicko and Yang Xiao*, Engineering Transport–Reaction Coupling in MXene-Supported Catalysts for Ethane Dehydrogenation, *2026 AIChE Annual Meeting, Catalysis and Reaction Engineer-*

ing Division, Minneapolis, MN, November 8–12, **2026**.

56. Tobias K. Misicko, Joaquin Herrero, Jacob Robinson, and Yang Xiao*, Generalized Criterion for Quantification of Chemical Reactor Runaway Risk, *2026 AIChE Annual Meeting, Catalysis and Reaction Engineering Division*, Minneapolis, MN, November 8–12, **2026**.
55. Tobias K. Misicko and Yang Xiao*, Methane Activation Via Selective Passivized Catalysis, *2026 AIChE Annual Meeting, Hydrocarbon Conversion I: Methane*, Minneapolis, MN, November 8–12, **2026**.
54. Yang Xiao*, Selective Passivized Catalysis: Methane as an In-Situ Hydrogen Donor, *Materials Seminar, James Tarpo Jr. and Margaret Tarpo Department of Chemistry, Purdue University*, West Lafayette, IN, August 28, **2026**.
53. Tobias K. Misicko and Yang J. Xiao*, Selective Passivized Catalysis: Bridging the Gap for Low-Temperature Methane Activation, *ACS Fall 2026 National Meeting, C1 Catalysis, Session II (CATL)*, Chicago, IL, August 26, **2026**.
52. Joaquin Herrero, Daniela S. Mainardi, and Yang Xiao*, Combating Climate Change: Upgrading CO_2 into Ethanol Using Cu/CeO_{2-x} Catalysts, *the Southwest Catalysis Society (SWCS) Spring Symposium*, Houston, TX, May 1, **2026**.
51. Jiaping Weng, Xiaoyang Gao, Daniela S. Mainardi*, and Yang Xiao*, Computational Design and Discovery of High-Entropy MXene for Stabilizing Confined Pt Nanolayer, *the Southwest Catalysis Society (SWCS) Spring Symposium*, Houston, TX, May 1, **2026**.
50. Keshar Sanjel, Tobias Misicko, Yang Xiao, and Daniela S. Mainardi*, Confinement-Enhanced Ethane Dehydrogenation on Pt Nanolayer/Mo_2TiC_2 MXene Catalysts: A DFT Study, *the Southwest Catalysis Society (SWCS) Spring Symposium*, Houston, TX, May 1, **2026**.
49. Tobias K. Misicko and Yang Xiao*, Selective Passivization of Pt Catalysts for Hydrogen Production, *the Southwest Catalysis Society (SWCS) Spring Symposium*, Houston, TX, May 1, **2026**.
48. Tobias K. Misicko and Yang Xiao*, Selective Passivized Catalysis for Direct Hydrogen Production from Methane, *the Southwest Catalysis Society (SWCS) Spring Symposium*, Houston, TX, May 1, **2026**.
47. Yang Xiao*, Selective Passivized Catalysis: Balancing Activity, Selectivity, and Stability in Methane Activation, ***invited talk for L. K. Doraiswamy Graduate Seminar, Department of Chemical and Biological Engineering, Iowa State University***, Ames, IA, February 26, **2026**.
46. Tobias Misicko, and Yang Xiao*, Unraveling the Synergistic Role of Cu(I) and Oxygen Vacancies in CO_2 Hydrogenative Coupling to Ethanol over Cu/CeO_{2-x} Catalysts, *the 14th Next-Generation Applications of Nanotechnology for Optimal Sustainability (NANO) Conference*, San Diego, CA, November 7 - 9, **2025**.
45. Tobias Misicko, and Yang Xiao*, Engineering Reaction-Diffusion Behavior in MXene-Based Catalysts for Ethane Dehydrogenation, *AIChE Annual Meeting*, Boston, MA, November 2 - 6, **2025**.
44. Tobias Misicko, XiaoPeng Liu, Yue Wu*, and Yang Xiao*, Activation of Short Alkanes via Selective Passivized Catalysis, *2025 Great Plains Catalysis Society Annual Symposium*, Ames, Iowa, August 10 - 11, **2025**.

43. Tobias Misicko and Yang Xiao, Quantifying Reaction-Diffusion Phenomena of Ethane Dehydrogenation on Two-Dimensional Metal Carbide Catalyst, *The 29th North American Catalysis Society Meeting (NAM29)*, Atlanta, Georgia, June 8 - 13, **2025**.
42. Yang Xiao, Methane Activation: Bridging the Gap Between Catalysis and Reaction Engineering ***(invited talk for graduate students in the department of chemical engineering)***, *University of Louisiana at Lafayette*, Lafayette, Louisiana, February 24, **2025**.
41. Joaquin Herrero, Tobias Misicko, and Yang Xiao, Machine Learning for Parametric Sensitivity of Chemical Reactors, *The 5th North American Symposium on Chemical Reaction Engineering (NASCRE-5)*, Houston, Texas, February 16 - 19, **2025**.
40. Tobias Misicko and Yang Xiao, Quantifying Reaction-Diffusion Rates of Nonoxidative Coupling of Methane per Active Edge Sites of Two-Dimensional Pt Nanolayer Catalysts, *The 5th North American Symposium on Chemical Reaction Engineering (NASCRE-5)*, Houston, Texas, February 16 - 19, **2025**.
39. Yang Xiao, Tobias Misicko, and Xiaoyang Gao, Thermal Runaway of Chemical Reactors: An Experimental, Modeling and Machine-Learning Investigation, *AIChE Annual Meeting*, San Diego, CA, October 27 - 31, **2024**.
38. Yang Xiao, Tobias Misicko, Xiaoyang Gao, and Daniela Mainardi, Catalytically Active Edge Sites of MXene-Confined Pt Nanolayers for Conversion of Short Alkanes, *AIChE Annual Meeting*, San Diego, CA, October 27 - 31, **2024**.
37. Yang Xiao, Zhe Li, Tobias Misicko, Jeff T. Miller, Yue Wu, and Arvind Varma, Atomically Thin Platinum Nanolayers on MXene for Catalytic Non-oxidative Coupling of Methane, *The 27th International Symposium for Chemical Reaction Engineering (ISCRE 27)*, Quebec City, Quebec, Canada, June 11 - 14, **2023**.
36. Tobias Misicko and Yang Xiao, Thermal Runaway of Catalytic Reactors: an Experimental, Modeling and Machine-Learning Study, *Southeastern Catalysis Society 2023 Annual Symposium*, Tuscaloosa, AL, February 27 - 28, **2023**.
35. Yang Xiao, Tobias Misicko, Jeffrey Miller, and Yue Wu, Atomically Thin Pt Nanolayer Catalysts Supported on Two-dimensional Molybdenum Titanium Carbide (MXene) for Shale Gas Conversion, *Southeastern Catalysis Society 2023 Annual Symposium*, Tuscaloosa, AL, February 27 - 28, **2023**.
34. Zhe Li, Yang Xiao, Prabudhya Chowdhury, Zhenwei Wu, Tao Ma, Johnny Zhuchen, Gang Wan, Tae-Hoon Kim, Dapeng Jing, Peilei He, Pratik Potdar, Lin Zhou, Zhenhua Zeng, Xiulin Ruan, Jeffrey T. Miller, Jeffrey Greeley, Yue Wu, and Arvind Varma, Nonoxidation Coupling of Methane over Nano-Layer Platinum Catalysts on Two-Dimensional Metal Carbides (MXenes), *AIChE Annual Meeting*, Phoenix, AZ, November 13 - 18, **2022**.
33. Yang Xiao, Anand Ramanathan, Bala Subramaniam, and Arvind Varma, Guaiacol Hydrodeoxygenation and Hydrogenation over Bimetallic Pt-M (Nb, W, Zr)/KIT-6 Catalysts with Tunable Acidity, *AIChE Annual Meeting*, Phoenix, AZ, November 13 - 18, **2022**.
32. Yang Xiao, Pratik Potdar, Kaida Liu, Arvind Varma, and Guomin Xiao, A Machine Learning Tool for Thermal Runaway Prediction of Chemical Reactors, *AIChE Annual Meeting* (Virtual), November 16 -

20, **2020**.

31. Yang Xiao, Feng Jiang, Guomin Xiao, and Arvind Varma, CO_2 Hydrogenation to Ethanol over Pd/Bi_2O_3 Catalysts: The Synergistic Effect of Pd Particle Size and Surface Oxygen Vacancy, *AIChE Annual Meeting* (Virtual), November 16 - 20, **2020**.
30. Yang Xiao and Arvind Varma, Parametric Sensitivity and Runaway in Fixed-Bed Reactors: Example of Methanol Selective Oxidation over Pt-Bi Catalysts, *AIChE Annual Meeting*, Orlando, Florida, November 10 - 15, **2019**.
29. Yang Xiao, Anand Ramanathan, Bala Subramaniam, and Arvind Varma, Enhanced Aromatic Selectivity during Deoxygenation of Phenolic Model Compounds over Bifunctional Catalysts, *AIChE Annual Meeting*, Orlando, Florida, November 10-15, **2019**.
28. Yang Xiao, Anand Ramanathan, Bala Subramaniam, and Arvind Varma, Enhanced Aromatic Selectivity during Deoxygenation of Phenolic Model Compounds over Bifunctional Catalysts, *2019 North American Catalysis Society Meeting (NAM26)*, Chicago, Illinois, June 23-28, **2019**.
27. Johnny ZhuChen, Zhenwei Wu, Slgi Choi, Yang Xiao, Arvind Varma, Wei Liu, Guanghui Zhang, and Jeffrey T. Miller, Structure Determination of Nanoscale Pt_3Bi Intermetallic Alloy Catalysts for Non-Oxidative Coupling of Methane and Propane Dehydrogenation, *2019 North American Catalysis Society Meeting (NAM26)*, Chicago, Illinois, June 23-28, **2019**.
26. Yang Xiao and Arvind Varma, A Mechanistic Study of Glycerol Conversion to Aromatic Hydrocarbons over Bifunctional Metal-Supported H-ZSM-5 Catalysts ***(invited talk in honor of Prof. Dan Luss' contributions to the field of chemical reaction engineering)***, *The 4th North American Symposium for Chemical Reaction Engineering (NASCRE 4)*, Houston, Texas, March 10-13, **2019**.
25. Yang Xiao and Arvind Varma, Highly Selective Nonoxidative Coupling of Methane (NOCM) over Pt-Bi Bimetallic Catalysts, *AIChE Annual Meeting*, Pittsburgh, Pennsylvania, October 28 - November 2, **2018**.
24. Arvind Varma and Yang Xiao, Bio-Oil Upgrading Using Methane: A Mechanistic Study of Model Compound Guaiacol Reactions over Pt-Bi Bimetallic Catalysts ***(invited talk in honor of Prof. Doraiswami Ramkrishna's contributions to the field of chemical reaction engineering)***, *AIChE Annual Meeting*, Pittsburgh, Pennsylvania, October 28 - November 2, **2018**.
23. Yang Xiao, Yuan Wang, and Arvind Varma, Methanol Conversion to Formaldehyde at Low Temperatures over Pt-Bi Bimetallic Catalysts, *AIChE Annual Meeting*, Pittsburgh, Pennsylvania, October 28 - November 2, **2018**.
22. Yang Xiao and Arvind Varma, Bimetallic Catalysis for Various Shale Gas and Biomass Conversions, *AIChE Annual Meeting*, Pittsburgh, Pennsylvania, October 28 - November 2, **2018**.
21. Rexonni B. Lagare, Yang Xiao, and Arvind Varma, Direct Catalytic Conversion of Methane to Liquid Oxygenates, *CISTAR Biannual Meeting*, Albuquerque, NM, October 3-5, **2018**.
20. Yang Xiao and Arvind Varma, Upgrading Bio-Oil Model Compound over Pt-Based Catalysts: A Comparative Study Using Hydrogen and Methane, *International Symposium on Chemical Reaction Engineering (ISCRE-25)*, Florence, Italy, May 20 - 23, **2018**.

19. Yang Xiao and Arvind Varma, Highly Selective Nonoxidative Coupling of Methane (NOCM) over Pt-Bi Bimetallic Catalysts, *The Catalysis Club of Chicago 2018 Spring Symposium*, Chicago, Illinois, May 11, **2018**.
18. Yang Xiao, Yuan Wang, and Arvind Varma, Low-Temperature Selective Oxidation of Methanol to Formaldehyde over Pt-Bi Bimetallic Catalysts, *39th Annual Michigan Catalysis Society Spring Symposium*, Midland, Michigan, May 3, **2018**.
17. Lindsey Blanshan, Yang Xiao, and Arvind Varma, Catalytic Conversion of Biomass Model Compounds to Biofuels over Pt-based Supported Catalysts, *AIChE North Central Regional Conference*, West Lafayette, IN, April 6 - 7, **2018**.
16. Yang Xiao and Arvind Varma, Insight and Applications of Pt-Bi Bimetallic Catalysts: A Combined Experimental and DFT Study, *AIChE Annual Meeting*, Minneapolis, MN, October 29 - November 3, **2017**.
15. Yang Xiao and Arvind Varma, Kinetics of Glycerol Conversion to Hydrocarbon Fuels over Pd/H-ZSM-5 Catalysts, *AIChE Annual Meeting*, Minneapolis, MN, October 29 - November 3, **2017**.
14. Yang Xiao and Arvind Varma, Kinetics of Guaiacol Deoxygenation Using Methane over Pt-Bi Catalyst, *25th North American Catalysis Society Meeting*, Denver, CO, June 4-9, **2017**.
13. Xiaohong Wang, Yang Xiao, and Arvind Varma, Controllable Solution Combustion Synthesis of Nanoscale α / $\beta - Bi_2O_3$ and Its Catalytic Application, *AIChE Annual Meeting*, San Francisco, CA, November 13-18, **2016**.
12. Yang Xiao and Arvind Varma, Glycerol to Hydrocarbon Fuels Via Bifunctional Catalysts, *AIChE Annual Meeting*, San Francisco, CA, November 13-18, **2016**.
11. Yang Xiao and Arvind Varma, Guaiacol Deoxygenation Using Methane over Pt-Bi Catalysts: Reaction Pathways and Kinetics, *AIChE Annual Meeting*, San Francisco, CA, November 13-18, **2016**.
10. Yang Xiao and Arvind Varma, Insight into Pt-Bi Bimetallic Catalysts: An Experimental and DFT Study, *AIChE Annual Meeting*, San Francisco, CA, November 13-18, **2016**.
9. Yang Xiao and Arvind Varma, Insight into Pt-Bi Bimetallic Catalysts for Tuning Selectivity and Improving Stability, *International Symposium on Chemical Reaction Engineering (ISCRE-24)*, Minneapolis, MN, June 12-15, **2016**.
8. Yang Xiao and Arvind Varma, Catalytic Deoxygenation of Guaiacol Using Methane, *AIChE Annual Meeting*, Salt Lake City, UT, November 8-13, **2015**.
7. Yang Xiao, Zhi-Jian Zhao, Jeffrey Greeley, and Arvind Varma, An Experimental and Theoretical Study of Glycerol Selective Oxidation to 1,3-Dihydroxyacetone Via Bimetallic Platinum-Bismuth Catalysts, *AIChE Annual Meeting*, Salt Lake City, UT, November 8-13, **2015**.
6. Yang Xiao, Zhi-Jian Zhao, Jeffrey Greeley, and Arvind Varma, Glycerol Selective Oxidation to 1,3-Dihydroxyacetone via Bimetallic Platinum-Bismuth Catalysts: An Experimental and Theoretical Study, *AIChE 7th Annual Midwest Regional Conference*, Chicago, IL, March 12-13, **2015**.
5. Yang Xiao, Guomin Xiao, and Arvind Varma, Experimental and Simulation Study of a Universal Proce-

dure for Crude Glycerol Purification From Different Feedstocks in Biodiesel Production, *AIChE Annual Meeting*, San Francisco, CA, November 3-8, **2013**.

4. Yang Xiao and Arvind Varma, Experimental and Simulation Study of Crude Glycerol Purification from Different Feedstocks in Biodiesel Production, *246th ACS National Meeting*, Indianapolis, IN, September 8-12, **2013**.
3. Yang Xiao, Haoyang Li, Guomin Xiao, Lijing Gao, and Xiaomei Pan, Simulation of the Catalytic Reactive Distillation Process for Biodiesel Production via Transesterification, *International Conference on Renewable Energy and Environmental Materials*, Beijing, China, May 21-23, **2012**.
2. Lijing Gao, Guomin Xiao, and Yang Xiao, Study on Intrinsic Kinetics of Transesterification Reaction Catalyzed by Solid Base in Fixed Bed Reactor, *The 6th National Conference on Chemical Engineering and Biochemistry*, Changsha, China, October 29-31, **2010**.
1. Yang Xiao, Lijing Gao, Guomin Xiao, and Jianhua Lv, Kinetics of Transesterification Reaction Catalyzed by Solid Base in Fixed Bed Reactor, *International Conference on Chemical and Biological Utilization of Biomass Resources (ICCUB2010)*, Nanjing, China, October 22-26, **2010**.

Professional Services

Service to Professional Society, Department, Research Center, College, and University

Secretary	Southwest Catalysis Society	2026 - 2028
Tenure-Track Faculty Search Committee	Chemical Engineering Department	2024 - 2025
Executive Committee	Institute for Micromanufacturing	2023 - 2024
Strategic Planning Committee - Research Team	College of Engineering and Science	2024 - 2025
University Senate (Executive Committee)	Louisiana Tech University	2024 - 2025
University Senate	Louisiana Tech University	2025 - 2028

Journal Editorships and Advisory Boards

Catalysts	Guest Editor	2021 - present
Int. J. Comput. Mater. Sci. Surf. Eng.	Acting Editor	2021 - present
Sustainability	Editorial Board Member	2022 - present

Proposal Panelist/Reviewer

Funding Agency:	*Program:*	
National Science Foundation (NSF)	CBET Catalysis	2023
National Science Foundation (NSF)	Major Research Instrumentation	2024
National Science Foundation (NSF)	Trailblazer Impact Award	2024
National Science Foundation (NSF)	Process Systems, Reaction Engineering, and Molecular Thermodynamics	2024
National Science Foundation (NSF)	Environmental Engineering	2025
ACS Petroleum Research Fund (PRF)	Surface Science	2023-2025
Department of Energy (DOE)	Office of Basic Energy Sciences	2023

Conference Sessions Chaired

9. Hydrocarbon Conversion (I: Methane; II: Light Alkanes; III: Olefins and Others) (bin chair), *AIChE Annual Meeting*, Minneapolis, MN, November 8 - 12, 2026.
8. C1 Catalysis (symposium organizer), *ACS Fall Meeting*, Chicago, IL, August 23 - 27, 2026.
7. Catalyst Design, Synthesis, and Characterization V: Hydrogenation/Dehydrogenation (session chair), *AIChE Annual Meeting*, Boston, MA, November 2 - 6, 2025.
6. Catalysis on Low Dimensional Materials (bin co-chair, session chair), *AIChE Annual Meeting*, San Diego, CA, October 27 - 31, 2024.
5. Catalysis on Low Dimensional Materials (bin co-chair, session co-chair), *AIChE Annual Meeting*, Orlando, FL, November 5 - 10, 2023.
4. Session III, *Southeastern Catalysis Society Annual Symposium*, Tuscaloosa, AL, February 27 - 28, 2023.
3. Fundamentals of Catalysis and Surface Science II: Zeolites and Acid Catalysis, *AIChE Annual Meeting*, Phoenix, AZ, November 13 - 18, 2022.
2. Biomass Upgrading I: Reaction Fundamentals, *AIChE Annual Meeting*, Boston, MA, November 5 - 11, 2021.
1. Fundamentals and Strategies for Catalytic Biomass Conversion, *AIChE Annual Meeting*, Virtual, November 16 - 20, 2020.

Manuscript Reviewer (200+ reviews for 40+ journals)

ACS Omega, *ACS Nano*, *ACS Sustainable Chemistry & Engineering*, *AIChE Journal*, *Applied Energy*, *Applied Catalysis A: General*, *Applied Surface Science*, *Asia-Pacific Journal of Chemical Engineering*, *Biomass and Bioenergy*, *Biomass Conversion and Biorefinery*, *Catalysis Communications*, *Catalysis Letters*, *Catalysts*, *Chem*, *Chem Catalysis*, *ChemNanoMat*, *Chemical Engineering and Processing: Process Intensification*, *Chemical Engineering Journal*, *Chemical Engineering Science*, *ChemistrySelect*, *ChemCatChem*, *ChemSusChem*, *Chinese Journal of Chemical Engineering*, *ECS Journal of Solid State Science and Technology*, *Energy Exploration & Exploitation*, *Energy & Fuels*, *Fuel Processing*, *Journal of Environmental Chemical Engineering*, *Journal of Sustainable Bioenergy Systems*, *Industrial & Engineering Chemistry Research*, *Korean Journal of Chemical Engineering*, *Materials*, *Materials Chemistry and Physics*, *Molecular Simulation*, *Molecular Physics*, *Molecular Catalysis*, *Molecules*, *Next Research*, *Reactions*, *Reaction Chemistry & Engineering*, *Results in Surfaces and Interfaces*, *Reviews in Chemical Engineering*, *Sustainability*, and *Sustainable Energy & Fuels* .